X measured, low noise

X + Y measured, low noise

X + Y measured, high noise

A

B

C

Concentration

2.1

1.4

0.7

0

0

15

30

45

time

X (True simulation)

X (Measurement)

X (Fit, known)

X (Fit, param)

X (Fit, UDE)

Y (True simulation)

Y (Measurement)

Y (Fit, known)

Y (Fit, param)

Y (Fit, UDE)

D

E

F

Max. likelihood

0

-2

-4

-1

0

1

log(d)

Profile (known)

Profile (param.)

Profile (UDE)

d (true)

95% conf. interval

G

H

I

A(Y;p)

0.9

0.6

0.3

0

0.45

0.9

1.35

Y

$A(Y;\boldsymbol{p}) = 1.6\frac{Y^4}{1.5^4 + Y^4}$

$A(Y;\boldsymbol{p}) = v\frac{Y^n}{K^n + Y^n}$

$A(Y;\boldsymbol{p}) = U(Y;\boldsymbol{\theta})$